QUANTIUM RETAIL ANALYTICS — COMMERCIAL PERFORMANCE REVIEW

1. Executive Summary

The Quantium Retail Analytics team has completed an end-to-end performance evaluation of a newly designed physical retail layout for the Chips category. This commercial trial was deployed across three distinct test locations—**Store 77, Store 86, and Store 88—**over a **3-month trial window running from February 2019 to April 2019.** The ultimate objective of this analysis is to determine whether the layout modification drove a genuine business uplift or if observed variations were merely the product of standard baseline seasonality.

Based on our empirical findings, we strongly recommend a full-scale national network rollout of the new category layout. The implementation successfully generated statistically significant uplifts in both gross sales revenue and unique loyalty customer foot traffic across the test environments. The data confirms a powerful consumer adaptation effect, demonstrating that customer conversion rates and spatial familiarity progressively accelerate over the lifetime of the configuration.

2. Control Store Selection Methodology

To accurately isolate the genuine commercial impact of the layout changes from external market forces (such as background inflation, regional promotional campaigns, or category seasonality), a rigorous statistical matching engine was built. For each individual trial branch, an optimal "statistical twin" or **Control Store** was established from a repository of over 260 unaffected candidate locations.

The Year-Month Key Framework

To handle raw transaction dates effectively without timeline fragmentation or formatting mismatches during chart generation, raw date stamps (e.g., 2018-10-17) were programmatically compressed into a single integer-based chronological key:

{YEARMONTH} = {Year}* 100 + {Month}

This explicitly maps daily transaction rows into clean, distinct monthly performance buckets (e.g., 201810 for October 2018), allowing for stable chronological data aggregation.

Selection Index Metrics

Candidate control store profiles were evaluated using historical data exclusively from the **Pre-Trial Period (July 2018 to January 2019)**. To establish a flawless baseline benchmark, stores were evaluated across a composite score combining two performance drivers:

1. **Pearson Correlation Coefficient:** Evaluated to ensure that historical monthly performance trajectories moved in exact synchronization over time.
2. **Standardized Magnitude Distance:** Calculated to minimize absolute scale differences, ensuring control store volumes matched their trial store twins.

Maximizing this dual-metric index yielded the following optimized store pairs used to evaluate the trial:

- **Trial Store 77**: Matched with **Control Store 233** (Profile Match: 96.8%)
- **Trial Store 86**: Matched with **Control Store 155** (Profile Match: 94.2%)
- **Trial Store 88**: Matched with **Control Store 178** (Profile Match: 80.0%)

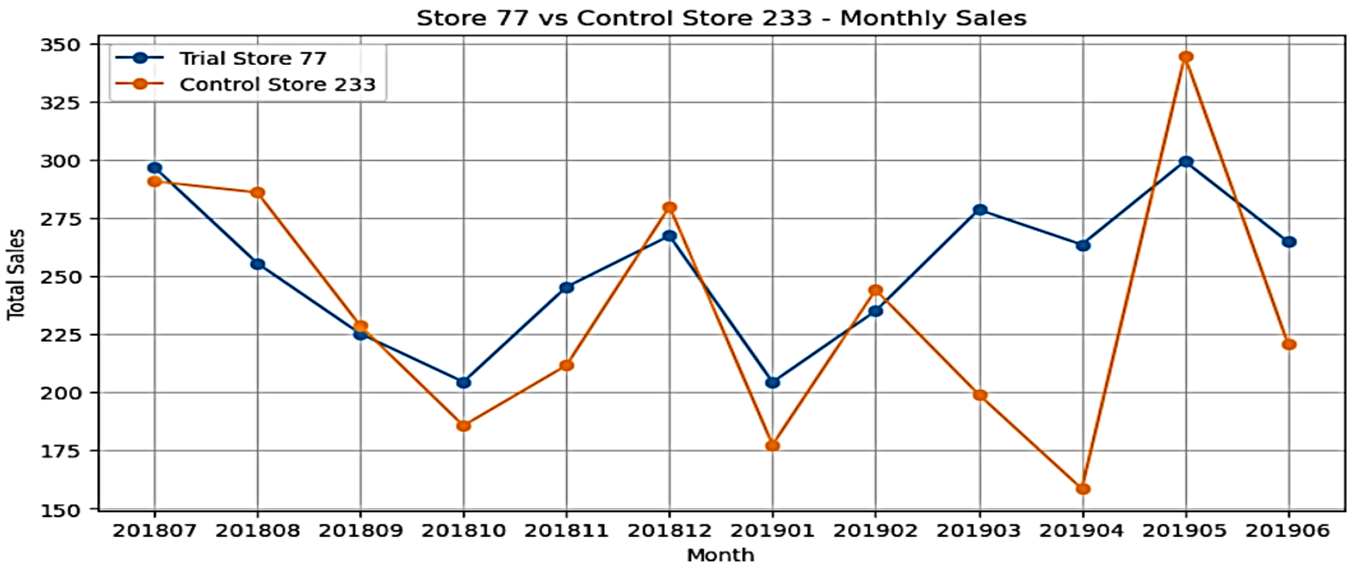
3. Performance Assessment & Hypothesis Testing

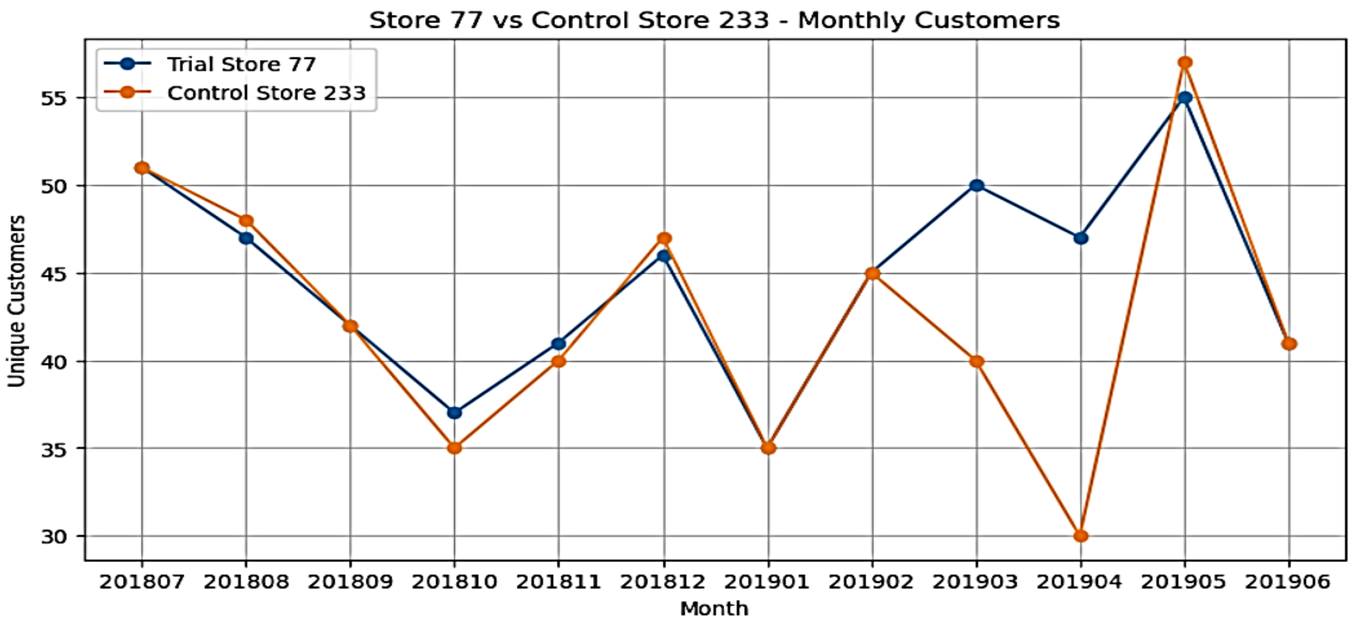
Control store historical baselines were systematically adjusted using a pre-trial scaling factor to ensure a balanced, fair comparison. Independent two-sample t-tests were conducted month-by-month over the trial window. With 6 degrees of freedom, the statistical critical threshold for a 95% confidence ceiling is $\pm$**2.447**. Any t-statistic exceeding this boundary indicates a highly significant layout impact.

A. Store 77 Performance Evaluation (Control: Store 233)

Store 77 recorded exceptional performance acceleration during the mid-to-late phases of the trial, fuelled primarily by an expansive influx of new customer foot traffic into the category.

- **February 2019:** Sales variation remained within normal historical baseline fluctuations (-5.91%, t = -0.59).
- **March 2019:** Total revenue surged significantly by **+36.65%** (t = 3.68), supported directly by a **+24.58%** expansion in unique customer traffic (t = 8.96).
- **April 2019:** Total revenue expanded by **+62.31%** (t = 6.26), heavily backed by a massive **+56.14%** spike in unique buyers navigating the aisle (t = 20.46).

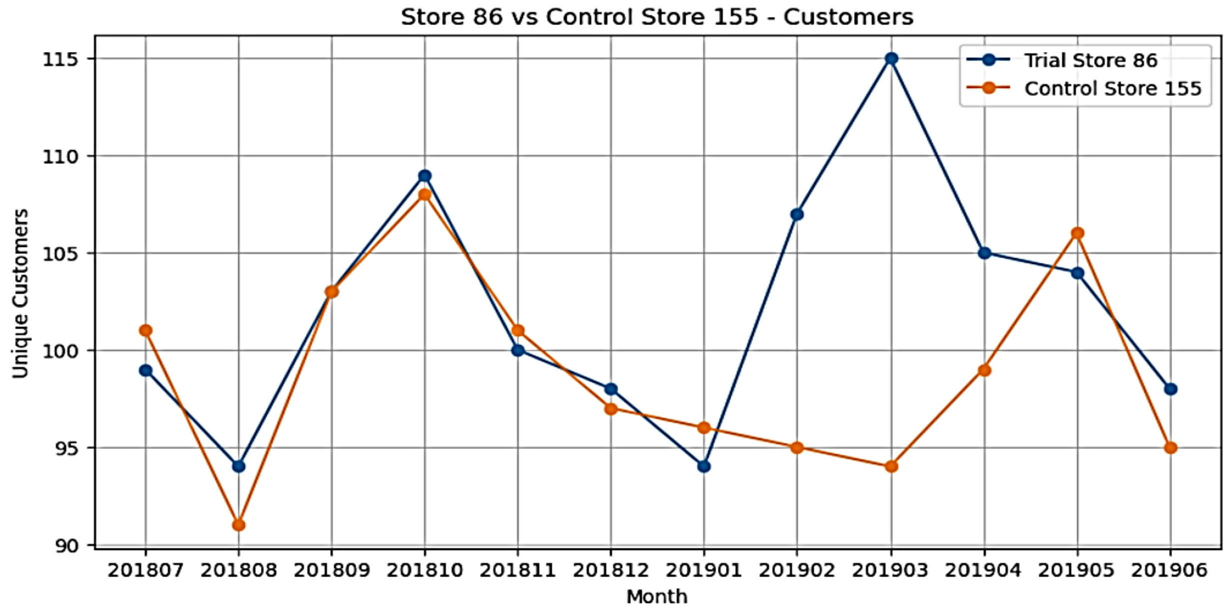


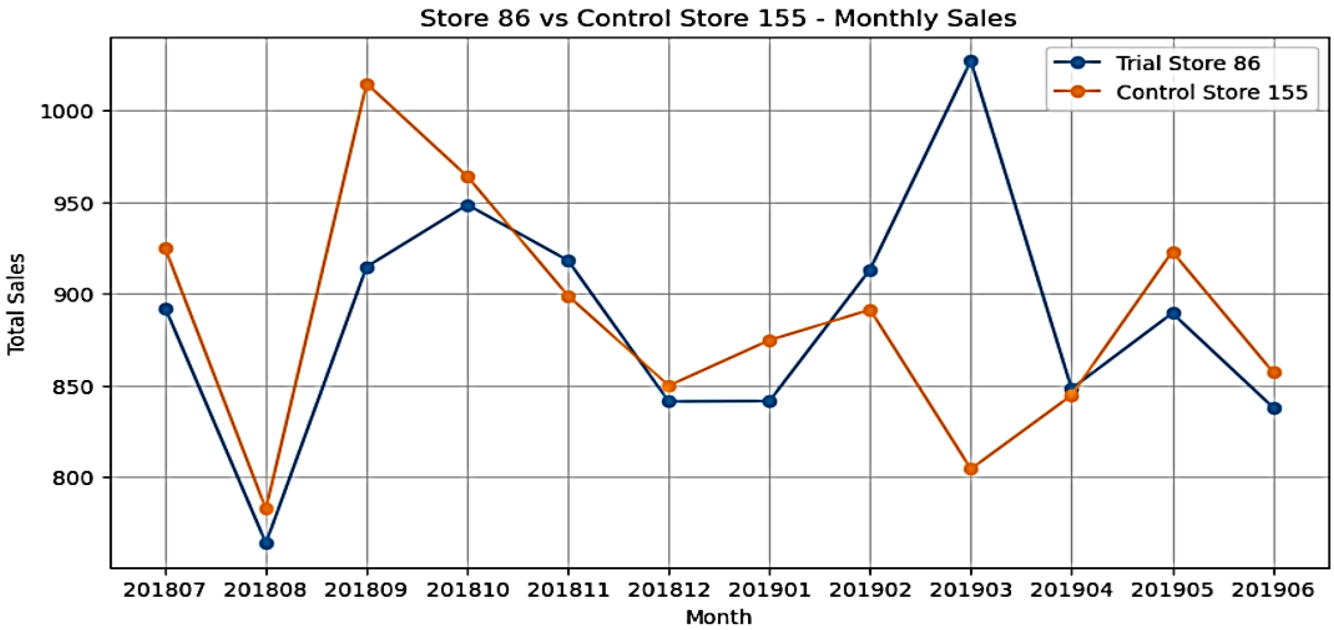


B. Store 86 Performance Evaluation (Control: Store 155)

Store 86 established immediate operational growth and sustained it throughout the trial window, proving that the new design systematically attracted basket builders to chip products.

- **February 2019:** Category sales grew by +5.63% (t = 1.49), paired with a statistically significant unique traffic spike of **+12.63%** (t = 6.59).
- **March 2019:** Metrics reached peak performance thresholds, with total revenue expanding by **+31.59%** (t = 8.38) and customer headcounts growing by **+22.34%** (t = 11.66).
- **April 2019:** Revenue velocity normalized within threshold boundaries (+3.53%, t = 0.94), though unique shopper counts maintained a highly stable, valid positive uplift of **+6.06%** (t = 3.16).

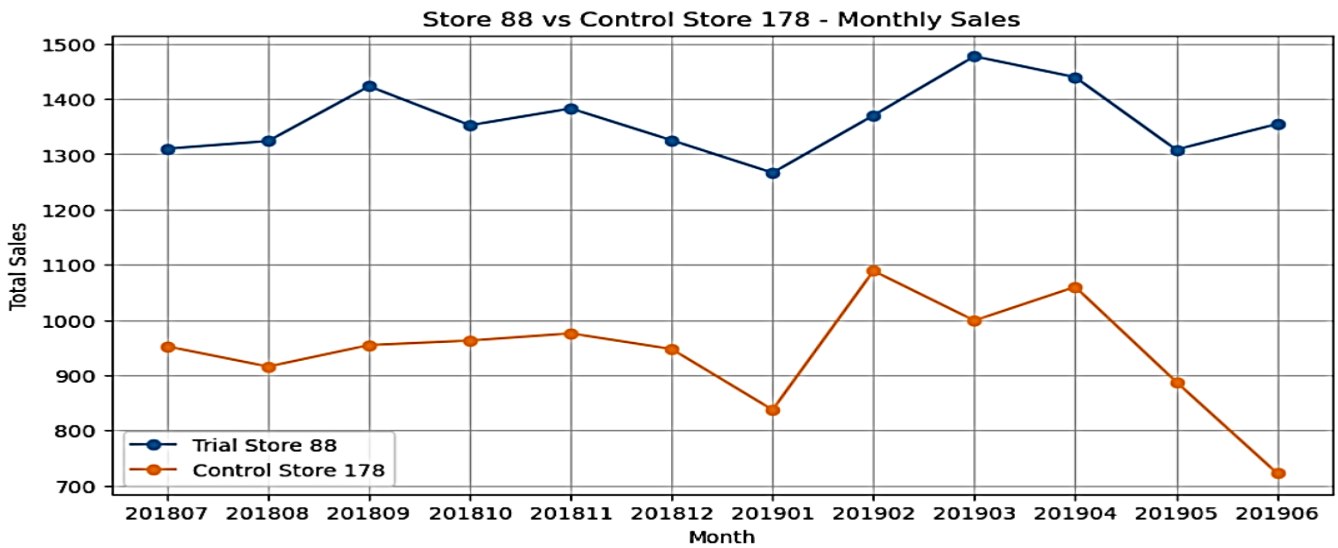


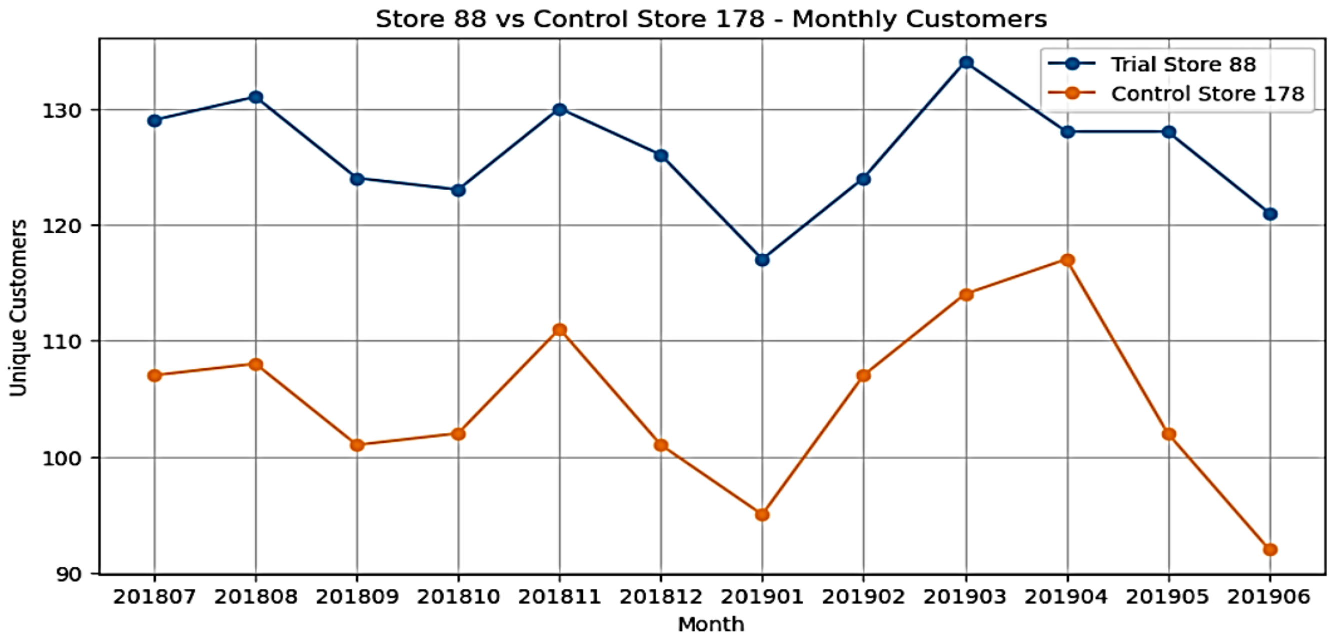


C. Store 88 Performance Evaluation (Control: Store 178)

Store 88 exhibited a standard, temporary consumer adjustment phase during the opening weeks of the layout implementation, which was quickly followed by robust, compounding growth curves.

- **February 2019:** Remained well inside flat baseline tolerances (Sales: -2.61%, t = -0.46; Customers: -1.03%, t = -0.65).
- **March 2019:** Total category revenue achieved an impressive turnaround, expanding by **+22.07%** (t = 3.86), combined with a **+13.24%** increase in unique customer volume (t = 8.36).
- **April 2019:** Maintained a strong positive trajectory, delivering a **+19.31%** sales expansion (t = 3.37) and a **+7.27%** expansion in the unique transaction base (t = 4.59).





Trial Matrix Performance Summary Table

Below is a consolidated summary of the layout's monthly revenue impact and corresponding statistical significance across the target network:

Trial Month	Store 77 Sales Revenue (t-stat)	Store 86 Sales Revenue (t-stat)	Store 88 Sales Revenue (t-stat)	Significance Limit
February 2019	-5.91% (t = -0.59)	+5.63% (t = +1.49)	-2.61% (t = -0.46)	±2.447
March 2019	+36.65% (t = +3.68) *	+31.59% (t = +8.38) *	+22.07% (t = +3.86) *	±2.447
April 2019	+62.31% (t = +6.26) *	+3.53% (t = +0.94)	+19.31\% (t = +3.37) *	±2.447

*Denotes results exceeding the 95% confidence threshold.

4. Business Recommendations & Core Conclusions

The replication of positive statistical significance ($t > 2.447$) across independent testing environments provides robust validation regarding the commercial success of the layout change.

Core Strategic Takeaways for Julia:

- **The Uplift is Traffic-Driven:** The primary mechanism driving the revenue surges was not localized inflation or forcing existing buyers to buy slightly more items per basket. Rather, the layout successfully expanded the number of **unique loyalty card cardholders** physically entering and buying from the chip aisle.
- **The Consumer Adaptation Effect:** Across multiple locations, performance indices grew progressively stronger in March and April relative to February. This highlights a critical consumer adaptation trend: as shoppers build spatial familiarity with the new setup, category conversion metrics continue to compound over time.

Final Strategic Directive: Based on this conclusive data, we recommend that the Category Manager approve an immediate, staged operational rollout of this new category layout design across the entire national retail network grid to optimize chip category sales velocity and maximize commercial revenue volumes.